Midterm 2 Practice — Lectures 12–21

M 375T Mathematical Foundations of Machine Learning — Fall 2026

Midterm Thursday, November 19, in class · 75 minutes · closed book, no devices

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NumPy you should be able to read:
`np.array`, `np.ones`, `np.eye`, `np.outer`, `np.sqrt`, `np.abs`, `np.round`, `np.sort`, `np.mean`, `np.sum(..., axis=1)`, `np.cumsum(..., axis=1)`, `x.var()`, `x.std()`, `rng.choice`, `W.sum(axis=1)`, `P.T @ mu`, `Q @ w`, `X.T @ X`, `np.linalg.norm`, `np.linalg.solve`, `np.linalg.eigvals`, `np.linalg.eigh` (eigenvalues of a symmetric matrix in increasing order, eigenvectors as columns), `np.linalg.svd` (singular values in decreasing order), boolean indexing `P[d > 0]`.
Conventions. Laws are column vectors, $\mu_{t+1} = P^{\mathsf{T}}\mu_t$. Data points are the rows of X ($n \times d$); the rows are centered and $S = X^{\mathsf{T}}X/n$. Eigenvalues of S are ordered $\lambda_1 \geq \lambda_2 \geq \cdots$, with orthonormal eigenvectors $q_1, q_2, \ldots$ `rng = np.random.default_rng(0)` throughout. Do everything on paper first, then check in NumPy.

Part A — PageRank and walks on directed graphs

A1. A directed graph from code. $W_{ij} = 1$ means there is a link from page i to page j.

```
W = np.array([[0, 1, 1, 0],
              [0, 0, 1, 0],
              [1, 0, 0, 1],
              [0, 0, 0, 0]], dtype=float)
d = W.sum(axis=1)
P = np.ones((4, 4)) / 4
P[d > 0] = W[d > 0] / d[d > 0, None]
print(P.sum(axis=1))              # line 1
mu = np.array([1., 0., 0., 0.])
for t in range(2):
    mu = P.T @ mu
print(mu)                         # line 2
```

(a) Draw the graph with its arrows. Write d. Which vertex is dangling, and why can the rule $P_{ij} = W_{ij}/d_i$ not be used for it?

(b) Write P. What does the code do with the dangling vertex? Predict line 1.

(c) Predict line 2.

(d) Is $\pi \propto d$ invariant here? Check that $\pi = (4, 3, 6, 4)/17$ satisfies $P^{\mathsf{T}}\pi = \pi$ at vertex 2. Vertex 2 has the largest π; which property of the graph explains this?

A2. Teleportation. P is the directed 3-cycle $0 \to 1 \to 2 \to 0$, and $G_\alpha = \alpha P + (1-\alpha)\frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ with $n = 3$.

(a) Describe one step of the surfer that G_α models. Write the equation $G_\alpha^{\mathsf{T}}\pi = \pi$ entry by entry (one equation for π_j).

(b) Write $G_{1/2}$. Show that the uniform vector is invariant for it.

(c) The eigenvalues of P are $1,\ e^{2\pi i/3},\ e^{-2\pi i/3}$. State the eigenvalues of G_α, and predict the output of

```
G = 0.5 * P + 0.5 * np.ones((3, 3)) / 3
print(np.sort(np.abs(np.linalg.eigvals(G))))
```

(d) The ordinary walk on the cycle started at vertex 0 never converges. For G_α we proved $\|\mu_t - \pi\|_1 \le 2\alpha^t$. How many steps guarantee $\|\mu_t - \pi\|_1 < 0.01$ when $\alpha = \frac{1}{2}$? When $\alpha = 0.85$? (Use $\log 200 \approx 5.30$, $\log(1/0.85) \approx 0.1625$.)

A3. Personalized PageRank by hand. Three pages with links $0 \to 1$, $1 \to 0$, $2 \to 0$. Take $\alpha = \frac{1}{2}$ and $G_v = \alpha P + (1-\alpha)\mathbf{1}v^{\mathsf{T}}$, so that $\pi_j = \alpha \sum_i \pi_i P_{ij} + (1-\alpha)v_j$.

(a) Write P. With the uniform $v = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, find π. Start with π_2: nobody links to page 2.

(b) Find π for $v = e_2$ (every restart goes to page 2) and for $v = e_0$. Which page gets rank zero for $v = e_0$, and why?

(c) Find π for $v = e_1$, and check that the answer of (a) is the average of the three personalized vectors. Which property of the equation gives this?

Part B — Sampling a given measure

B1. A sped-up walk from code.

```
M, n = 10_000, 400
steps = rng.choice([-1., 1.], size=(M, n)) / np.sqrt(n)
pos = np.cumsum(steps, axis=1)
print(pos.shape)                                  # line 1
print(pos[:, -1].var())                           # line 2
print(pos[:, 99].var())                           # line 3
print(np.mean(np.abs(pos[:, -1]) > 1.96))         # line 4
```

(a) Predict line 1. What is one row of `pos`? What is the column `pos[:, -1]`?

(b) Predict lines 2 and 3 (to two decimals). Which time t does column 99 correspond to?

(c) Predict line 4 (to two decimals). Which theorem do you use?

(d) Brownian motion has independent increments. Given $B_1 = 2$, what is the distribution of B_3?

B2. Equilibrium and detailed balance.

(a) Write the equation that says π is invariant for P, and the equation that says (π, P) is in detailed balance. Say in one sentence what each one balances.

(b) Prove that detailed balance implies invariance.

(c) Undirected triangle with weights $W_{01} = 1$, $W_{12} = 2$, $W_{02} = 1$ and $P_{ij} = W_{ij}/d_i$. Find d, π, and check detailed balance on each of the three edges.

(d) Let $R = \begin{pmatrix} 0 & \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & 0 & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} & 0 \end{pmatrix}$ and $\pi = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. Is π invariant for R? Is (π, R) in detailed balance? Starting from π, how much mass moves $0 \to 1$ in one step, and how much moves $1 \to 0$?

B3. Metropolis by hand. States $0, 1, 2$ on a line, target $\pi = (0.2, 0.3, 0.5)$. Proposal: from the current state propose the left or the right neighbour with probability $\frac{1}{2}$ each; a proposal that leaves $\{0, 1, 2\}$ is refused. A proposal $i \to j$ is accepted with probability $\min(1, \pi_j/\pi_i)$; otherwise the walker stays, and staying counts as a step.

(a) Write the 3×3 transition matrix P. Check that every row sums to 1.

(b) Verify $P^{\mathsf{T}}\pi = \pi$ at every state, and detailed balance on the pair $(1, 2)$.

(c) The walker is at state 1, the proposal is state 0, and the uniform random number is $U = 0.8$. Where is the walker after this step? Same question with $U = 0.5$.

(d) A run of 50 000 steps counts the visits to each state. What do the three fractions approach? The algorithm uses only the ratios π_j/π_i; why is that useful?

Part C — Vectors, orthonormal bases, and projection

C1. Energy, cosine, and orthonormal coordinates.

(a) Let $u = (1, 2, 2)$, $v = (2, -2, 1)$, $w = (2, 1, 2)$. Compute $\|u\|^2$, $\|v\|^2$, $u \cdot v$ and $\|u + v\|^2$. Expand $\|u + v\|^2$ in general and say when the energies add. Compute $\cos(u, w)$.

(b) Predict the three outputs. Which identity does line 2 check? What does line 3 show?

```
Q = np.array([[1, 1], [1, -1]]) / np.sqrt(2)
w = np.array([5., 1.])
c = Q @ w
print(c * np.sqrt(2))              # line 1
print(c @ c)                       # line 2
print(Q.T @ c)                     # line 3
```

C2. The Haar basis of $\mathbb{R}^8$. The first four Haar vectors are

$$q_0 = \tfrac{1}{\sqrt{8}}(1,1,1,1,1,1,1,1), \qquad q_1 = \tfrac{1}{\sqrt{8}}(1,1,1,1,-1,-1,-1,-1),$$

$$q_2 = \tfrac{1}{2}(1,1,-1,-1,0,0,0,0), \qquad q_3 = \tfrac{1}{2}(0,0,0,0,1,1,-1,-1),$$

and $q_4, \dots, q_7$ are $\frac{1}{\sqrt{2}}(1, -1, 0, \dots, 0)$ and its shifts by two positions. Let $x = (6, 6, 2, 2, 1, 1, 1, 1)$.

(a) Write $x = \sum_k c_k q_k$: find the eight numbers c_k. Check Parseval.

(b) Compute the projection of x onto the span of q_0; of q_0, q_1; of $q_0, \dots, q_3$. Give the energy of the error each time, and say which coordinates it is made of.

(c) Let ξ have independent entries of mean 0 and variance σ^2, and let q be a unit vector. Show $\mathbb{E}[(q \cdot \xi)^2] = \sigma^2$. How much noise energy survives, on average, when $x + \xi$ is projected onto the span of $q_0, \dots, q_3$?

C3. The closest point in a span.

(a) Let $q_1, \dots, q_m$ be orthonormal and $E(c) = \|x - \sum_k c_k q_k\|^2$. Expand $E(c)$ and show that the minimizer is $c_k = q_k \cdot x$. Why is this critical point the global minimum?

(b) $q_1 = \frac{1}{3}(1, 2, 2)$, $q_2 = \frac{1}{3}(2, 1, -2)$, $x = (3, 3, 3)$. Check that q_1, q_2 are orthonormal. Compute the closest point y in their span, the residual $r = x - y$, and check $r \cdot q_1 = r \cdot q_2 = 0$ and $\|x\|^2 = \|y\|^2 + \|r\|^2$. Give $\cos(x, y)$.

(c) The vectors $a_1 = (1, 1, 0)$, $a_2 = (0, 1, 1)$ are not orthogonal. Write the normal equations $A^\mathsf{T}Ac = A^\mathsf{T}x$ for $x = (1, 0, 1)$, solve them, and give the closest point $y = Ac$ and the residual. Which line of code computes c?

C4. Random directions are almost orthogonal.

```
M, N = 5000, 400
U = rng.choice([-1., 1.], size=(M, N))
V = rng.choice([-1., 1.], size=(M, N))
cos = (U * V).sum(axis=1) / N
print(cos.shape)                  # line 1
print(cos.mean())                 # line 2
print(cos.std())                  # line 3
```

(a) Why is `(U * V).sum(axis=1) / N` the cosine between row i of `U` and row i of `V`?

(b) Predict the three lines (lines 2 and 3 to two decimals) and justify line 3.

(c) Two images of the same digit in $\mathbb{R}^{784}$ have cosine about 0.5 after removing the mean image. How many standard deviations of a random cosine is that? What does it say about the data?

Part D — Quadratic energies, diagonalization, and PCA

D1. Remove the cross term. $S = \begin{pmatrix} 3 & 2 \\ 2 & 6 \end{pmatrix}$ and $E(x) = x^\mathsf{T}Sx = 3a^2 + 4ab + 6b^2$ for $x = (a, b)$.

(a) Find the eigenvalues $\lambda_1 \geq \lambda_2$ and orthonormal eigenvectors q_1, q_2.

(b) With $u = q_1 \cdot x$ and $v = q_2 \cdot x$, write E in terms of u, v and verify it by expanding.

(c) Over unit vectors x, what are the largest and the smallest value of E, and where are they attained?

(d) A centered data set has covariance S. Describe the ellipse $x^\mathsf{T}S^{-1}x = 1$: directions and lengths of its half-axes.

(e) Predict the output.

```
lam, Q = np.linalg.eigh(S)
print(lam)
print(np.round(Q.T @ S @ Q, 6))
```

D2. PCA on four points. The data are $x_1 = (2, 6)$, $x_2 = (-2, -6)$, $x_3 = (3, -1)$, $x_4 = (-3, 1)$.

(a) We look for a unit vector q and numbers $c_1, \ldots, c_4$ minimizing $\frac{1}{4}\sum_i \|x_i - c_i q\|^2$. For a fixed q, find the best c_i. Then show that the error equals $\frac{1}{4}\sum_i \|x_i\|^2 - q^\mathsf{T}Sq$, and conclude which q is best.

(b) Compute S, its eigenvalues and eigenvectors. (Hint: compute Sx_1 and Sx_3.) Give the best q, the four c_i, the minimal error, and the share of the variance captured.

(c) The maximum-likelihood Gaussian for these data has mean 0 and which covariance? Give the major axis of its ellipse and the standard deviation along it.

(d) Predict the four outputs.

```
X = np.array([[2., 6.], [-2., -6.], [3., -1.], [-3., 1.]])
S = X.T @ X / 4
lam, Q = np.linalg.eigh(S)
q = Q[:, -1]
c = X @ q
Xhat = np.outer(c, q)
print(lam)                                          # line 1
print(np.round(np.abs(c), 3))                       # line 2
print(np.mean(np.sum((X - Xhat)**2, axis=1)))       # line 3
print(np.linalg.svd(X)[1]**2 / 4)                   # line 4
```

(You may use $\sqrt{10} \approx 3.162$.) Why does line 2 need `np.abs`?

Part E — The SVD, best approximation, and the power method

E1. An SVD by hand. Let $Y = \begin{pmatrix} 2 & 2 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}$.

(a) Compute $Y^{\mathsf{T}}Y$, its eigenvalues and eigenvectors. Give $\sigma_1 \geq \sigma_2$, v_1, v_2, and $u_k = Yv_k/\sigma_k$. Check that u_1, u_2 are orthonormal.

(b) Write $Y = \sigma_1 u_1 v_1^{\mathsf{T}} + \sigma_2 u_2 v_2^{\mathsf{T}}$ with both pieces as explicit matrices.

(c) Keep only the first piece Y_1. Add the squares of the entries of $Y - Y_1$ and compare with σ_2^2.

(d) State the Eckart–Young theorem for $k = 1$. Can a rank-one matrix B have $\|Y - B\|_F^2 = 3$?

E2. The SVD of a data matrix. X is $n \times d$ with centered rows, $n = 100$, $d = 7$, and $X = U\Sigma V^{\mathsf{T}}$ is the thin SVD.

(a) Give the shapes of U, Σ, V^{T}, $X^{\mathsf{T}}X$ and XX^{T}. Which of the last two is n times the covariance matrix?

(b) Suppose $X^{\mathsf{T}}Xq = 900\,q$ with $\|q\| = 1$. Find $\|Xq\|$, the singular value, and the eigenvalue of S. Show that $u = Xq/\|Xq\|$ is an eigenvector of XX^{T} and give its eigenvalue.

(c) The eigenvalues of S are $9, 4, 1, 0, 0, 0, 0$. Give the singular values of X. If every point is replaced by its projection onto the best line through the origin, what is the average squared error? Onto the best plane? Can any other plane do better?

E3. The power method. $S = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$ has eigenvalues 6 and 1 with $q_1 = (2,1)/\sqrt{5}$, $q_2 = (1,-2)/\sqrt{5}$.

(a) Start from $x_0 = (1, 0)$ and do two steps $x \leftarrow Sx$ (do not normalize; directions are what matter). Compute the Rayleigh quotient $R(x) = x^{\mathsf{T}}Sx/x^{\mathsf{T}}x$ at x_0, x_1, x_2.

(b) Write $x_0 = c_1 q_1 + c_2 q_2$. The tangent of the angle between x_t and q_1 is $|c_2\lambda_2^t|/|c_1\lambda_1^t|$. Give it for $t = 0, 1, 2$. By which factor does it shrink per step? By which factor does the gap $6 - R(x_t)$ shrink?

(c) What happens if $x_0 = q_2$ exactly? What happens in floating-point arithmetic if x_0 is q_2 plus a tiny perturbation?

(d) For a data matrix the step is coded as below. Why is S never formed? Compare the number of multiplications of one step with the cost of forming S, for $n = 12\,000$ and $d = 784$.

```
x = X.T @ (X @ x) / n
x = x / np.linalg.norm(x)
```

Part F — Nonlinear features, overfitting, and gradient descent

F1. More features, smaller training error. Training points (x, y): $(-1, 1)$, $(0, 0)$, $(1, 1)$.

(a) Fit $\hat{y} = c_0 + c_1 x$ by least squares: write Φ, the normal equations, and solve. Give the training error $\frac{1}{3}\sum_i (\hat{y}_i - y_i)^2$.

(b) Fit $\hat{y} = c_0 + c_1 x + c_2 x^2$. Give the coefficients and the training error.

(c) Explain why adding a feature can never increase the training error.

(d) A test point is $(2, 2)$. Compute the squared test error of each model. Which quantity should choose between the two models, and what is this phenomenon called?

F2. A disk inside a ring. Class -1: $(0, 0)$, $(\frac{1}{2}, 0)$, $(0, -\frac{1}{2})$. Class $+1$: $(2, 0)$, $(-2, 0)$, $(0, 2)$, $(0, -2)$. A classifier predicts the sign of $\hat{y}$.

(a) Show that no linear model $\hat{y} = w \cdot x + b$ classifies all seven points correctly. (Use the points $(2, 0)$, $(-2, 0)$ and $(0, 0)$.)

(b) Use the features $\varphi(x) = (1,\ x_1^2 + x_2^2)$ and $\hat{y} = c_0 + c_1(x_1^2 + x_2^2)$ with $c = (-1, \frac{1}{2})$. Check all seven points. Describe the decision boundary.

(c) The model in (b) is called linear. Linear in what? In what is it not linear?

F3. Gradient descent on a parabola and on a bowl.

(a) $E(c) = c^2$ and $c \leftarrow c - hE'(c)$. Write the update as $c \leftarrow (\dots)\, c$. Starting from $c = 1$, give c after three steps for $h = 0.25$, $h = 0.75$, $h = 1.25$. For which h does the iteration converge?

(b) $E(c) = \frac{1}{2}(9c_1^2 + c_2^2)$. Predict the output, then repeat with `h = 0.25`.

```
lam = np.array([9., 1.])
c = np.array([1., 1.])
h = 0.2
for t in range(2):
    c = c - h * lam * c
print(c)
```

(c) State the stability condition for a quadratic energy with curvatures $\lambda_{\max} \geq \cdots \geq \lambda_{\min} > 0$. Which curvature limits the step size and which one sets the speed? For the least-squares energy $E(c) = \frac{1}{n}\|y - \Phi c\|^2$ write the gradient, and say which matrix the curvatures are eigenvalues of.

F4. One trained feature. The model is $\hat{y} = c\tanh(wx + b)$ and the energy on one data point is $E = (\hat{y} - y)^2$. Recall $\tanh'(s) = 1 - \tanh^2(s)$ and $\tanh(0) = 0$.

(a) Write $\partial E/\partial c$, $\partial E/\partial w$, $\partial E/\partial b$ by the chain rule.

(b) Evaluate them at $c = 1$, $w = 0$, $b = 0$ with the data point $(x, y) = (2, 1)$. Do one gradient step with $h = 0.1$.

(c) For the neuron $\hat{y}_i = \tanh(w \cdot x_i + b)$ on images x_i with labels $y_i = \pm 1$, the energy is $E = \frac{1}{n}\sum_i (\hat{y}_i - y_i)^2$. Why are there no normal equations for (w, b)? Starting from $w = 0$, $b = 0$, show that the first gradient step moves w in the direction $\sum_i y_i x_i$, and describe this vector in terms of the two class means.

F5. Sharpness, the stability threshold, and noise. $E(a, b) = \frac{1}{2}(ab - 1)^2$.

(a) Compute ∇E. Describe the set of minima.

(b) At a minimum the Hessian is $\begin{pmatrix} b^2 & ab \\ ab & a^2 \end{pmatrix}$. Find its two eigenvalues. The largest one is the sharpness; compute it at $(1, 1)$, $(2, \frac{1}{2})$ and $(3, \frac{1}{3})$.

(c) Gradient descent with step h can stay near a minimum only if the sharpness is below $2/h$. For $h = 0.4$, which of the three minima can it stay near? What does a run started near the others do?

(d) Do one gradient step from $(a, b) = (2, 1)$ with $h = 0.1$. Give E before and after.

(e) Stochastic gradient descent replaces the average over n data points by an average over a random batch of B. Is the batch gradient right on average? With $n = 12\,000$ and $d = 785$ parameters, compare the cost of one step for $B = n$ and for $B = 100$. How does the size of the noise around a minimum depend on h and on B?

Bring questions to office hours (Tue/Thu 5–6 PM, PMA 9.156).